1. 奇异性问题 & 结构优化

🇨🇳 中文（口语版）

我们先讲一下机构设计这部分。

在我们之前的设计中，其实存在一个比较严重的问题，就是并联奇异性。对于并联结构来说，这是一个很典型、也比较危险的问题。

简单来说，所谓并联奇异，就是在某些姿态下，即使所有电机都锁住了，动平台仍然可能获得额外的自由度，也就是说它会“失控”。这种情况下，机构会失去刚性，无法抵抗外力，在康复场景下其实是有安全隐患的。

我们一开始的处理方法是比较标准的，就是通过对系统的约束雅可比，也就是 A 矩阵进行求解，去找它的行列式或者秩什么时候退化，从而定位奇异点。然后我们基于这个结果去计算工作空间，再通过机械限位，比如限制舵机在 50 到 80 度之间，来避免进入这些奇异区域。

但是这样的方法其实有一个问题，就是我们只是“绕开”奇异点，而不是从结构上解决它。

所以我们上周调研了一篇论文，它给了我们一个很有启发的思路。它是通过 Adams 和 Ansys 做仿真，分析了不同结构参数对奇异性的影响，发现对于 3-RRR 机构来说，连杆长度比例和平台半径是最敏感的两个参数。

其中一个关键结论是，当靠近平台的连杆 l2l_2l2 是靠近基座连杆 l1l_1l1 的大约 1.15 倍时，整个机构的各向同性会明显提升，同时奇异曲线会被“推”到工作空间的边界。

换句话说，就是原来在工作空间内部的奇异点，被移到了边缘，甚至变成可控区域。

基于这个结论，我们重新优化了机构参数，并重新计算了奇异空间。结果我们发现：

- 整体各向同性提升了大约 20%
- 原本在工作空间内部的一些奇异区域基本被压缩到了边界
- 有效可用的安全工作空间明显变大了

所以我们现在不是单纯通过限位去“避开问题”，而是从结构上去“减少问题”。

🇺🇸 **English**（口语版）

Let me first talk about the mechanism design.

In our previous design, we encountered a significant issue: **parallel singularities**, which are quite common in parallel mechanisms.

A parallel singularity happens when the mobile platform gains uncontrolled degrees of freedom even when all actuators are locked. In other words, the system loses stiffness and becomes uncontrollable, which is especially dangerous in rehabilitation scenarios.

Initially, we handled this in a conventional way. We analyzed the constraint Jacobian, or the A-matrix, and identified singular configurations by checking when its determinant or rank degenerates. Then we computed the workspace and applied mechanical joint limits, like work between 30 to 80 degrees, to avoid those regions.

However, this approach only avoids singularities rather than solving them fundamentally.

Last month, we came across a paper that provided a more insightful solution. Using Adams and Ansys simulations, the study showed that for a 3-RRR mechanism, **link length ratio and platform radius are the most sensitive parameters affecting singularity distribution**.

Specifically, when the distal link l2 is about **1.15 times** the proximal link l1, the mechanism achieves better isotropy, and singularity curves are pushed toward the boundary of the workspace.

This essentially transforms previously problematic regions into controllable ones.

Based on this, we optimized our structure and recalculated the singularity distribution. We observed:

About a 20% improvement in isotropy

- **Internal singular regions were pushed toward the boundaries**
- **The usable and safe workspace significantly increased**

So instead of just avoiding singularities, we are now reducing them at the structural level.

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2. 康复策略（AAN）

🇨🇳 中文（口语版）

接下来讲一下我们的康复策略。

我们对踝关节康复领域做了比较系统的调研，最后选定了一套适合可穿戴设备的策略。

我们的设备不仅可以像 CPM 一样做被动康复，还可以支持主动康复。

为了更好地促进神经重塑，我们采用了 David Reinkensmeyer 提出的 **AAN**，也就是按需辅助策略。

和传统被动训练不一样，AAN 的核心是：
机器人不会一直带着你动，而是只在你“跟不上”的时候才提供帮助。

我们具体是通过 DYNAMIXEL 舵机的电流反馈，去实时估算用户的主动输出力矩 τext\tau_{ext}τext。当系统检测到用户已经无法跟踪目标轨迹时，才会通过动力学模型提供最小必要的辅助。

这样可以避免用户“偷懒”，同时也能强制大脑参与运动学习。

🇺🇸 English（口语版）

Next, I’ll talk about our rehabilitation strategy.

We conducted a thorough study of ankle rehabilitation methods and selected a strategy suitable for wearable devices.

Our system supports both passive rehabilitation, like CPM, and active rehabilitation.

To promote neuroplasticity, we adopted the **Assist-As-Needed (AAN)** strategy.

Unlike passive approaches, AAN only provides assistance when the user cannot keep up with the desired trajectory.

We estimate the user’s torque using motor current feedback. Assistance is provided only when necessary, ensuring active participation.

主动康复实验设计

我们设计了三类实验来量化效果：

第一类是模拟功能障碍实验，比如在脚尖加沙袋，或者通过舵机施加一个向下力矩，模拟足下垂。
我们观察开启 AAN 之后，LSI 对称性是否能从 80% 提升到 95%，以及提胯这种代偿动作是否减少。

第二类是系统透明度测试。
我们会让机器人进入一个接近零阻抗的模式，让用户穿着设备行走，看它是否影响自然步态。如果 IMU 测得的运动曲线相关性 R2R^2R2 能达到 0.95 以上，同时交互力矩接近 0，就说明系统几乎是“透明”的。

第三类是主动参与度测试。
我们会观察在 AAN 模式下，用户的主动出力占比是否逐渐提升，同时比较 jerk 平滑度，验证系统既能保持平滑性，又能提高响应速度。

被动康复实验

对于被动康复，我们主要验证步态迁移效应。

简单来说，就是让用户在设备上做一段时间训练，比如强化背屈，然后再下地走路，看是否会出现短期的步态变化，比如步幅增加、CoP 前移等。

如果这些变化显著存在，就说明我们的系统确实在影响神经控制策略。

Active Rehabilitation Experiment Design

For active rehabilitation, we designed three types of experiments to quantitatively evaluate the performance of our system.
We mainly use the following metrics for evaluation: LSI (Limb Symmetry Index), Jerk (motion smoothness), interaction torque, and human contribution ratio.

Our initial test subjects are the three of us in the team. Since our ankles are healthy, we need to artificially create an impairment and then use our device to perform rehabilitation testing.

First, simulated impairment.

We artificially create a condition similar to foot drop, either by adding a small weight to the toe, or by applying a downward torque using the motor.

Then we turn on the AAN mode and evaluate:

- whether the **LSI symmetry improves**, for example from around 80% to 95%,
- and whether **compensatory motions like hip hiking are reduced**

So this experiment basically checks if our system can **correct abnormal gait**.

Second, system transparency.

We set the robot to a near **zero-impedance mode**,
so it just follows the user without adding any assistance.

Then we compare:

- the **IMU signals** between barefoot walking and wearing the robot
- and also the **interaction torque**

If the motion similarity R squared is above 0.95,
 and the interaction torque is close to zero,

then it means the robot is almost **transparent**, so it doesn't interfere with natural walking.

Third, active participation.

Here we evaluate whether the user is actually engaging under AAN.

We mainly look at:

- the **human contribution ratio**, meaning how much effort comes from the user
- and the **jerk**, to make sure the motion is still smooth

So this tells us whether AAN can **encourage participation without making motion unstable**.

🎤 Passive Rehabilitation

For passive rehab, we focus on the **transfer effect**.

The idea is simple:

- the subject walks normally first
- then trains on the robot for a short time, for example with enhanced dorsiflexion
- and then walks again

We check:

- whether **step length increases**,
- and whether the **CoP（centre of pressure）shifts forward**

If we observe these short-term changes,
 it means the system is actually influencing the **user's neural control**.

✅ 一句总结（建议你讲 ）

Overall, we plan to use LSI, Jerk, interaction torque, CoP, and human contribution ratio as our key quantitative metrics to evaluate both rehabilitation effectiveness and human-robot interaction quality.

3. 传感器与系统设计

🇨🇳 中文（口语版）

基于实验室提供的舵机和gait mat，以及上述提到的几个量化指标进行传感器和主控制器的选型。我们最终敲定了主控选择树莓派和rs485 hat，方便和已有舵机通信，然后还需要两个imu传感器，一个固定在患者小腿（胫骨侧），一个固定在动平台或足部背侧。舵机编码器仅测量相对关节角，IMU 则提供相对于重力向量的绝对姿态。

另外，目前我们3D打印的结构还比较粗糙，佩戴体验一般。后续我们会加入鞋垫和软材料填充，来提升舒适性。

🇺🇸 **English**（口语版）

According to the servos and gait mat provided by the lab, and the metrics we discussed earlier, we selected our sensors and main controller.

We finally chose a Raspberry Pi with an RS485 HAT as the main controller, which allows convenient communication with the existing servo motors. In addition, we use two IMU sensors: one mounted on the patient's shank (tibia side), and the other attached to the moving platform or the dorsum of the foot.

While the servo encoders can only measure relative joint angles, the IMUs provide absolute orientation with respect to the gravity vector.

Currently, our 3D-printed structure is still relatively rough, and the wearing experience is not very comfortable. So this week, we plan to order and add an insole and soft padding to improve comfort.

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4. Milestone（7周计划）

🇨🇳 中文（口语版）

最后讲一下我们接下来 7 周的计划。

我们现在是三个人的团队。

- 第1周：确定传感器和主控方案，优化结构，确定 AAN 策略
- 第2周：完成底层控制和传感器标定
- 第3周：部署 AAN 算法并解析 gait 数据

- 第4到6周：做实验、采集数据、分析并优化策略
- 第7周：整理结果并完成论文

🇺🇸 English（口语版）

Finally, our 7-week plan:

- Week 1: finalize hardware and strategy
- Week 2: control and calibration
- Week 3: deploy AAN
- Week 4–6: experiments and optimization
- Week 7: paper writing